Practical # 04

Objective:

Implement and test different string operations such as concatenation, substring, comparison, and reversal.

Theory:

A string is a sequence of characters terminated by a special null character \0. Strings are widely used to process textual data such as names, messages, and commands. In C++, strings can be represented using:

- Character arrays (char str[ ])
- string class from the C++ Standard Library

In this lab, students will:

- Declare and initialize strings.
- Traverse characters in a string.
- Apply basic operations such as concatenation, substring extraction, comparison, and reversal.
- Analyse outputs for correctness.

C++ program: Write C++ to insert the strings at the end of string1.

```cpp
#include<iostream>
#include<cstring>
using namespace std;
int main()
{
    string str1,str2,str3;
    cout<<"Enter first string";
    cin>>str1;
    cout<<"Enter second string";
    cin>>str2;
    str3=str1+str2;
    cout<<str3;
    return 0;
}
```

OUTPUT

```
Enter first string Compter
Enter second string System
CompterSystem
```

Review Questions/ Exercise:

1. **Implement** a program to input and display a string.
2. **Develop** a program to concatenate two strings.
3. **Implement** a program to extract a substring from a given string.
4. **Develop** a program to compare two strings (equal, greater, or smaller).
5. **Construct** a program to reverse a string.
6. **Implement** a program to count the total number of vowels and consonants in a string.
7. **Develop** a program to check whether a given string is a palindrome.
8. **Construct** a program to find the length of a string without using the built-in length() function.
9. **Implement** a program to convert a string into uppercase and lowercase.
10. **Develop** a program to replace all spaces in a string with an underscore _.

Name: ________________________________

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